Predicting Polling Results of German parties with Google Trends Data of Political Topics

Jan Thiele

Motivation: Political science research has explored many mechanisms explaining voting behavior. One of such mechanisms are issue salience and issue ownership. Issue voting theory states that individuals vote for parties they associate with a particular issue and furthermore think these parties are most competent in dealing with (Dennison, 2019). According to this theory political parties' "own" specific issues they are primarily associated with. For example the German party "Die Grünen" is primarily associated with environmental and climate issues and the party AfD with issues concerning Immigration (Reinbold et al., 2025). Issue salience in itself can be described as a certain prominence an issue has for a voter or how important an issue is for them (Dennison, 2019). If an issue reaches a certain prominence or importance within an individual they are likely to gather information about said topic possibly influencing their voting decision in the end (Mellon, 2014). Therefore, monitoring such information search behavior for salient political issues could predict polling results. Traditionally issue salience is measured with "most important problem" questions (MIP questions) in political surveys (Dennison, 2019). We try to use google trends to measure this behavior. Google trends data are particularly useful, because they alleviate some weaknesses of traditional surveys. They are available over a long period of time at regular intervals and can be scaled to national and international levels (Mellon, 2014). This is particularly useful because issue salience can change over time making short term changes easier to uncover (Lorenz et al., 2022). Secondly, traditional survey methods show a discrepancies between what voters concern themselves with and they think politics should be about (Dennison, 2019). Google trends data on the other shows what voters are interested in without the shortcomings of traditional survey methodology like social desirability (Lorenz et al., 2022). Furthermore Mellon (2014) argues that aggregated issue salience results are more accurate in predicting voting outcomes over individual MIP and voting intent responses. For all these reasons we explore the following research question in this projecting: Does issue salience measured by google trends data predict polling results of political parties in Germany?

Background: Lorenz et al. (2022) used google trends data of immigration-related search terms to correlate with the polling results of the German party AfD. Based on group-threat theory they hypothesize that individuals perceiving immigrants as an outgroup threat are more likely to gather information about threats of immigration. They collected data on search terms associated with such perceived threats and correlated their search volume with polling results of the AfD. They used the AfD as an example because of their right leaning anti-immigrant stands in the German party system. They computed simple bivariate correlations, real-time correlations and temporary lagged correlations from 5th October 2013 until 5th October 2018 of weekly data. Their results showed that terms associated with a security threat of immigration like ("criminal refugees" or "criminal immigrants") were significantly positive related to polling results of the AfD. Temporary lagged

correlation increased up until 35 to 50 weeks of lag with security threat terms reaching its peak with r=0.5 around 35 weeks. While these results are promising they have their limitations as argued by the authors. For one they argued that no search term was consistently predicative of future trends. Additionally, they worried about the representative of the sample used in with google trends data. Lastly, they had concerns about the validity of their measurement.

Lorenz et al. (2022) explained the relationship of information search behavior with polling results by group-threat theory allowing them to investigate different facets of possible perceived threats of immigration. While this theoretical approach works for the issue of immigration it is harder to justify for other relevant political issues. Issues concerning climate change could be explained using the same mechanism but fail for issues such economic growth or social policies, which are all issues concerning voters and parties in the German political landscape (Reinbold et al., 2025). Considering these limitations, we try to explain the relationship between information search and voting behavior by issue voting and ownership theory. However, like Lorenz et al. (2022) we require a validation strategy to answer the question: Do google trends data measure issue salience?

Mellon (2014) developed a framework for researchers to validate internet search data as a measure for issue salience. He concentrated on content and criterion validity. Content validity was assessed by inspecting the top 50 related queries of a potential search terms and deciding if it covers the contents of a political issue sufficiently. Criterion validity was assessed by investigating the relationship between search indices and "most important problem"- questions of political surveys. This way he validated the search terms "economy", "immigration" and "terrorism" as appropriate measures of issue salience. We will use Mellon's (2014) framework to validate the google search indices of political salient issues.

We have to find issues concerning voters. Reinbold et al. (2025) examined voters issue salience during the German federal election of 2025. Their results indicated that voters of the AfD were primarily concerned with immigration related issues and voters of the BSW with Foreign and defense policies. Voters of the CDU/CSU and the FDP were mostly concerned about economic issues and those of "Die Grünen" worried about the environment. Lastly LINKE and SDP were mostly concerned with social justice and equality. These results give us a clear selection of potential google search terms we can validate with Mellon's (2014) approach.

Data: We retrieved in total four datasets for our analysis. Like Lorenz el al. (2022) we used polling results by FORSA, because of their weekly granularity aligning with the weekly format of google trends data. We retrieved the data from (https://www.wahlrecht.de/umfragen/forsa.htm) with the pandas library in python. We limited the time frame of this data to start and end of our google trends time series ranging from 13th September of 2020 to 14th of September 2025, because weekly google trends is only available for the past five year. Lastly, we computed averages for each week of the year.

We retrieved weekly google trends data from the last five years in Germany of the following topics: "human migration", "climate change", "military policy", "social policy", "economic growth" and

"inflation". Topics in google trends are a collection of search terms and synonyms pertaining a similar concept making them more holistic than simple search terms but are also less reliable (Glimpse, n.d.). To alleviate this downside, we checked the top 10 related queries to verify they cover the desired political issues. We retrieved each topic individually which indexes each search index on an individual scale from 0 to 100. Additionally, to the topics we retrieved google trends data of 6 immigration related search terms ("Immigration", "Einwanderung", "Zuwanderung", "Migration", "Asyl", "Flüchtlinge")[1] to investigate their relationship to polling results of the AfD similar to Lorenz et al. (2022). We applied the same preprocessing to these terms as with the topics. For validation we acquired data from the "ZDF-Politbarometer", a biweekly to monthly survey by the "Forschungsgruppe Wahlen" (Forschungsgruppe Wahlen, n.d.) on the political situation in Germany including "most important problem" questions. We accessed the data through the GESIS website (GESIS, n.d.). We only downloaded yearly datasets of 2020 until 2024, because 2025 accumulated dataset was not available yet. We extracted variables of interest mainly the two "most important problem" questions. Unfortunately, the surveys don't use a consistent terminology for political issues, therefore we had to summarize synonyms and similar wording. To this end we used the fuzzy matching library in python. After matching we calculated the percentage of participants listing a certain issue for each week of a year. Any answer which was not primarily associated with a particular party we decided to exclude in the following manner. Participant could name the most important and the second most important problem. Where participants did not answer with one of the topics we used the answer of the second most important problem. In the end we concentrated our analysis on the responses pertaining the following problems: "climate", "energy", "immigration", "economy", "social equality", "Ukrainian war" and "peace keeping efforts".

Methods: We validate our google search data with the framework of Mellon (2014). First, we inspect the top 10 related queries of each index to verify content validity. This step was already done during the data collection. Secondly, we remove seasonality from each time series to avoid correlations-based on seasonal repeating patterns. The search indices and MIP series have to match in their stationarity to be used in OLS regression to avoid spurious correlations. We assess stationarity by running an Augmented Dicker-Fuller Test. Where orders of integration are different, we either differentiate the search index and check regression residuals for stationarity or exclude the search term if the corresponding MIP series is non-stationary. All valid search indices are used as a predictor in an OLS regression onto the thematically corresponding MIP series. The quality of an index as a measure of issue salience is assessed by the R2.

Once validated we analyze the relationship between the search indices and the polling results by computing Pearson correlations between them and fitting and OLS regression between issues and their associated parties. Similarly to the validation analysis we account for seasonality, stationarity and autocorrelation by running the OLS regression with Newey-West standard error. Again, the

[1] English translation of search terms: Immigration, Emigration, Migration, Asylum, Refugees

performance of these models is measured by the R2. Lastly, we fit autoregressive models to the polling time series of the AfD and include search terms as exogenous variable to assess if these offer predictable value beyond the autocorrelations. We assess the performance of this model by comparing the Bayesian Information criterion.

Results of Validation Analysis: We removed seasonality from all-time series data on a weekly basis and interpolated missing data. For the MIP series this approach introduces a lot of synthetic data, because the data is mostly biweekly or monthly by nature. Therefore, we repeated the analysis concerning this time series on a monthly level, averaging values for each month of a year. On a weekly level the google topics "immigration", "military policy", "social policy" and "inflation" qualified for OLS regression because of the match of integration with their corresponding MIP series. We checked the residuals of "military policy" and "inflation" for stationarity, which was not significant disqualifying them as valid predictors. The OLS regression of "social policy" and MIP series "social equality" was not significant. The model with the topic "immigration" and MIP series "immigration" was significant (p-value= 0.0076; R2 = 0.1116). However, monthly data without interpolation showed no significant results. The same is true for all the other topics. Validating the immigration related search terms was done in the same manner. Only two search terms were significant: "Migration" (p-value= 0.0327; R2=0.07955) and "Zuwanderung" (p-value<0.001; R2 = 0.13426). On a monthly basis only "Zuwanderung was significant (p-value= 0.0172; R2 = 0.05478). Therefore, according to Mellon (2014) none of the search topics and terms qualify as appropriate measurements of issue salience.

Results of polling data analysis: In line with issue voting theory, we analyzed the relationship between polling results and the google trends data matching parties to their most associated topic according to Reinbold et al. (2025). We also analyzed correlation between MIP and polling results for comparison. AfD, CDU/CSU and FDP were significantly correlated with their respective topics of immigration, inflation and economy (Table 1). AfD polling results were significantly positive related to all search terms except for "Flüchtlinge" which showed a significant negative correlation (Table 2). CDU was significantly positive related with MIP responses of "economy" and FDP significantly negative with the MIP responses for "inflation". All other parties did not show a significant correlation to their respective topics (Table 3).

Table 1. Pearson Correlations of weekly party polling results and weekly google trends topics

	CDU	SPD	GRÜNE	FDP	LINKE	AfD
Human migration	0.02	-0.26***	-0.44***	-0.35***	-0.04	0.53***
Climate Change	-0.1	0.23***	0.04	0.08	-0.1	-0.1
Military Policy	-0.3***	0.59***	0.64***	0.52***	-0.14*	-0.55***
Social Policy	0.01	0.15	0.22***	0.15*	-0.02	-0.21***
Economic Growth	0.02	-0.2**	-0.34***	-0.26***	-0.15*	0.3***
Inflation	-0.36***	0.57***	0.36***	0.16*	-0.41***	-0.26***

Note. Designation for p-values: p<0.05 (*); p<0.01 (**); p<0.001 (***)

Table 2. Pearson Correlations of AfD polling results and immigration related search terms

	1.	2.	3.	4.	5.	6.	7.
1. AfD polling	1						
2. Immigration	0.77***	1					
3. Flüchtlinge	-0.18**	0.08	1				
4. Migration	0.62***	0.71***	-0.01	1			
5. Einwanderung	0.52***	0.66***	0.13*	0.68***	1		
6. Asyl	0.41***	0.55***	0.31***	0.79***	0.64***	1	
7. Zuwanderung	0.38***	0.48***	-0.02	0.78***	0.75***	0.69***	1

Note. Designation for p-values: p<0.05 (*); p<0.01 (**); p<0.001 (***)

Table 3. Pearson Correlations between parties' polling results and percentage of MIP answers (monthly)

	CDU/CSU	SPD	GRÜNE	FDP	LINKE	AfD
Climate	-0.42*	0.23	-0.1	0.82***	0.14	-0.33
Energy	-0.28	0.35*	0.18	-0.34	-0.67***	0.2
Immigration	0.35	-0.4*	-0.07	-0.18	0.2	0.17
Economy	0.5**	-0.65**	-0.21	-0.08	0.62**	0.17
Inflation	-0.22	0.29	0.11	-0.43*	-0.67***	0.34
Ukrainian war	-0.18	0.51**	0	-0.33	-0.57***	0.03
Peacekeeping	0.25	-0.37*	0.18*	0.43	0.69***	-0.43*
Social equality	-0.34	0.23	-0.12	0.61***	0.09	-0.23

Note. Designation for p-values: p<0.05 (*); p<0.01 (**); p<0.001 (***)

We performed OLS regression between parties and google trends topics. We checked for stationarity and differentiated variables were necessary. Variables which were both non-stationary, were regressed and then their residuals checked for stationarity. After this procedure only the model with inflation as a predictor and CDU/CSU polling results as dependent variable emerged as valid and significant (p-value<0.001; R2= 0.13074). All other models were either non-significant or did not have stationary residuals. We performed the same procedure with the immigration related search terms and the polling results of the AfD. Only "Immigration" and "Einwanderung" emerged as significant (p-value<0.001, R2=0.58987; p-value<0.001, R2=0.27397). However, their models' residuals were not stationary and after differencing both search indices and the polling series, the results were not significant. This means that while the search terms moved together with the polling results, they were not the cause of change within them. Similar to the topics and search terms, none of the MIP question were significant predictors of the parties polling results they are primarily associated with.

Figure 1 shows three autoregressive models differing in exogenous variables and number of lags. However, none of the with immigration search term data combined models performed better than

the simple lagged model as measured by BIC as seen in Table 4. This means that the exogenous variable did not show any predictive value beyond autocorrelations. Additionally, since the inflation topic was the only search indices significant in our OLS regression analysis, we fitted autoregressive model as the exogenous variable to assess its predictive value beyond simple autocorrelation. Figure 2 shows the four autoregressive models of predicting the CDU polling results. Similar to the AfD polling models none of the, with google trends data combined models showed a better performance than the simple lagged version based on BIC.

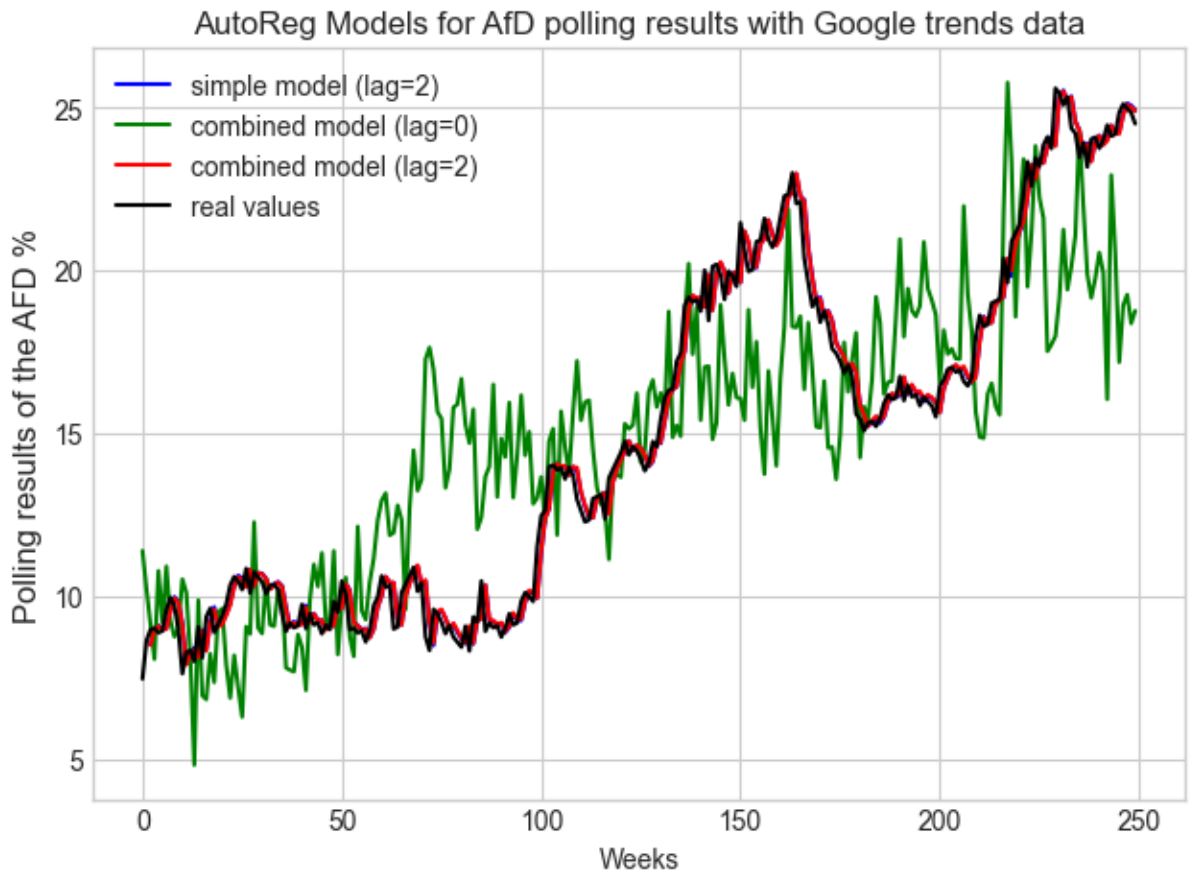


Figure 1. Autoregressive models of weekly AfD polling results with and without Google trends data of immigration related search terms

Table 4. Bayesian Information Criterion of Autoregressive

Models	BIC of AfD models	BIC of CDU/CSU models
Simple model with lag	475.251	710.765
Combined model without lag	1343.912	1310.781
Combined model with lag	485.425	716.147

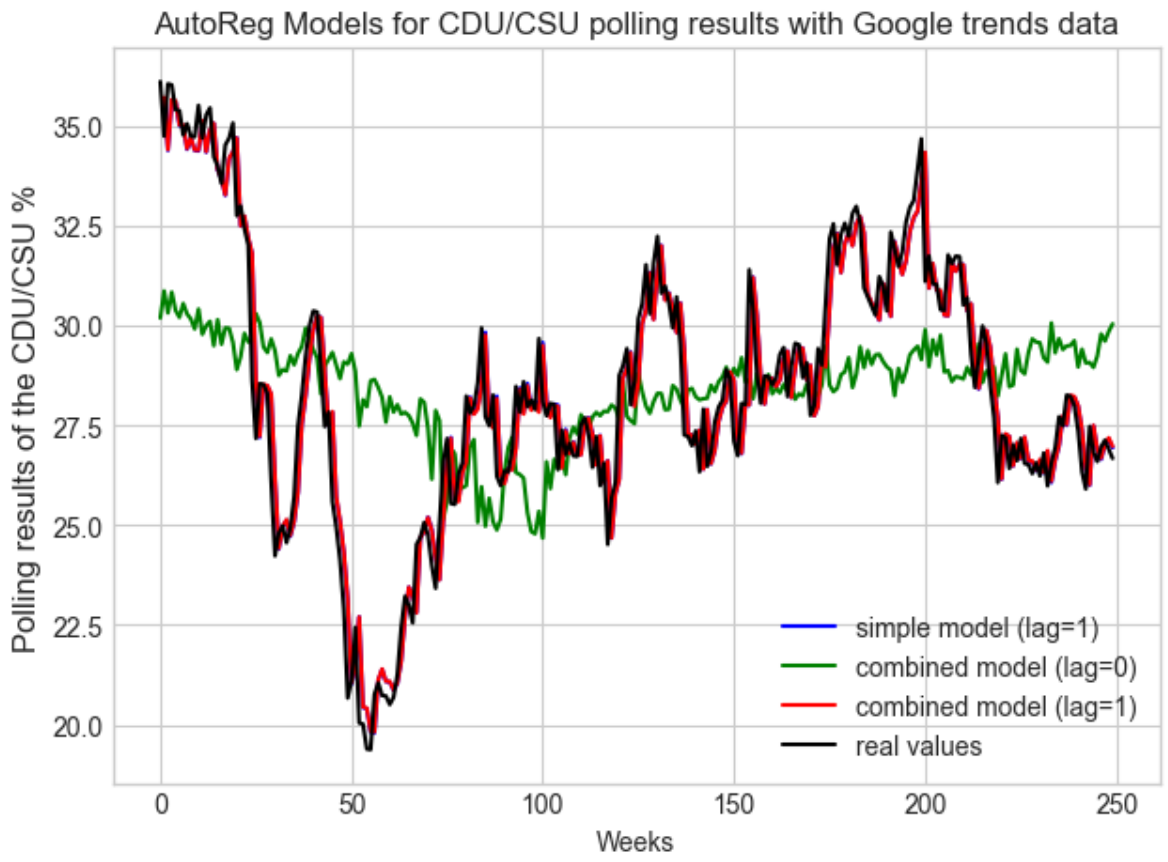


Figure 2. Autoregressive Models of weekly CDU/CSU polling results with and without Google Trends Data on the topic of inflation

Discussion: Our findings show that some parties polling results are correlated with their primarily associated topics namely CDU/CSU, FDP and AfD. Closer inspection of AfD revealed high positive correlations with immigration related search terms. On the other hand, aside from CDU/CSU and FDP no party showed significant correlations with MIP responses to the issues they "own". OLS regressions of topics showed only the topic "inflation" was significantly predictive of CDU/CSU polling results. Other topics, immigration related search terms and MIP responses were not significantly predictive of any polling results. Autoregressive Models with Google trends data as additional variables showed no predictive value of search data beyond autocorrelations. We found similar results to Lorenz et al. (2022) with the high correlations.

These findings suggests that while google trends are correlated with some parties, they are not predictive of their change. Therefore, we didn't find enough evidence to support our research question that issue salient topics measured by google trends data are predictive of polling results of German parties.

However, it seems that issue salience itself might not be a sufficient predictor of polling results because our analysis with MIP data did not yield significant results either, even for terms previously validated by Mellon (2014). Another explanation is that we did not measure issue

salience with our analysis. Possibly, we measured public attention of certain topics which do not necessarily influence polling results.

The results of this project have to be interpreted in the light of several limitations. During preprocessing of the MIP responses, we excluded the topic Coronavirus from our analysis which was highly prevalent during 2020 till 2022. Additionally, we didn’t have any control over representativeness of our sample. FORSA does it’s surveys mostly over the phone which might be a different sample than the average internet user.

References

Dennison, J. (2019). A Review of Public Issue Salience: Concepts, Determinants and Effects on Voting. *Political Studies Review*. https://doi.org/10.1177/1478929918819264

Forschungsgruppe Wahlen. (n.d.). *Forschungsgruppe Wahlen > Startseite*. Retrieved September 21, 2025, from https://www.forschungsgruppe.de/Startseite/

GESIS. (n.d.). *GESIS - Leibniz-Institut für Sozialwissenschaften*. Retrieved September 21, 2025, from https://www.gesis.org/wahlen/politbarometer

Glimpse. (n.d.). *Search Terms vs. Topics in Google Trends, Understand the differences between Google's two types of queries: Search Terms and Topics*. Glimpse. Retrieved September 20, 2025, from https://meetglimpse.com/google-trends/search-term-vs-topic/

Lorenz, R., Beck, J., Horneber, S., Keusch, F., & Antoun, C. (2022). Google Trends as a Tool for Public Opinion Research: An Illustration of the Perceived Threats of Immigration. In S. Pötzschke & S. Rinken (Eds.), *Migration Research in a Digitized World: Using Innovative Technology to Tackle Methodological Challenges* (pp. 193–206). Springer International Publishing. https://doi.org/10.1007/978-3-031-01319-5_10

Mellon, J. (2014). Internet Search Data and Issue Salience: The Properties of Google Trends as a Measure of Issue Salience. *Journal of Elections, Public Opinion and Parties*, *24*(1), 45–72. https://doi.org/10.1080/17457289.2013.846346

Reinbold, J., Kurz, K. R., Schleehauf, R., Fröhle, M., & Jäckle, S. (2025). Campaigning for whom? *Zeitschrift Für Vergleichende Politikwissenschaft*, *19*(3), 361–379. https://doi.org/10.1007/s12286-025-00644-z